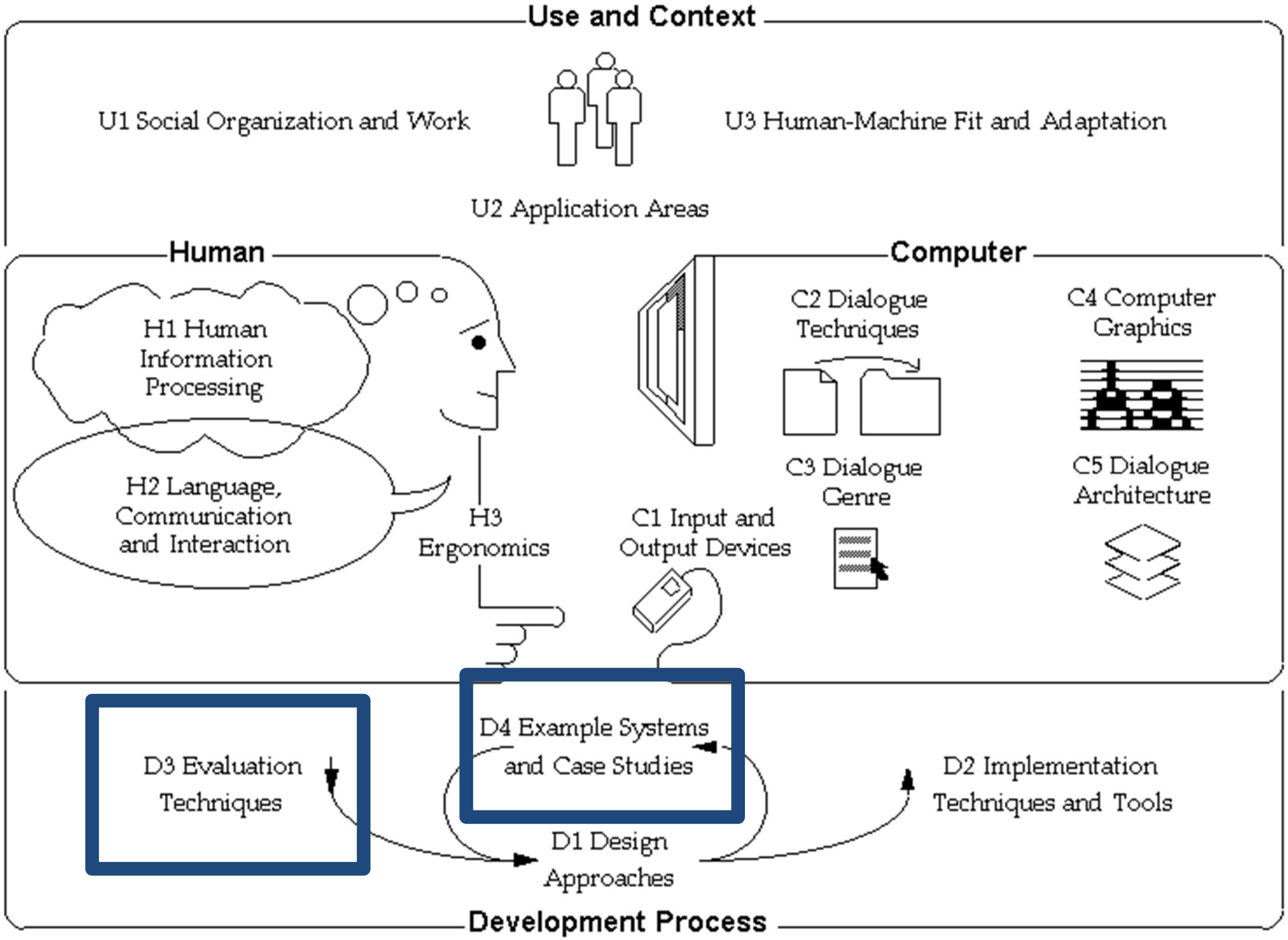
Human Computer Interaction

Johan F. HOORN

Lecture 06, Part 2: Evaluation techniques and sample systems

C-3PO ("3PO" aka Threepio)
Lukas, G., & Hales, J. (2002). *Star Wars Episode II: Attack of the Clones.* Los Angeles, CA: 20th Century Fox. [Part 7/10] - 1080p HD, 7:00 min. Retrieved May 21, 2012 from http://www.youtube.com/watch?v=-o5Y_AszZGg

Overview


Use and Context
U1 Social Organization and Work
U3 Human-Machine Fit and Adaptation
U2 Application Areas
Human
Computer
H1 Human Information Processing
H2 Language, Communication and Interaction
H3 Ergonomics
C1 Input and Output Devices
C2 Dialogue Techniques
C3 Dialogue Genre
C4 Computer Graphics
C5 Dialogue Architecture
D3 Evaluation Techniques
D4 Example Systems and Case Studies
D2 Implementation Techniques and Tools
D1 Design Approaches
Development Process

Contents

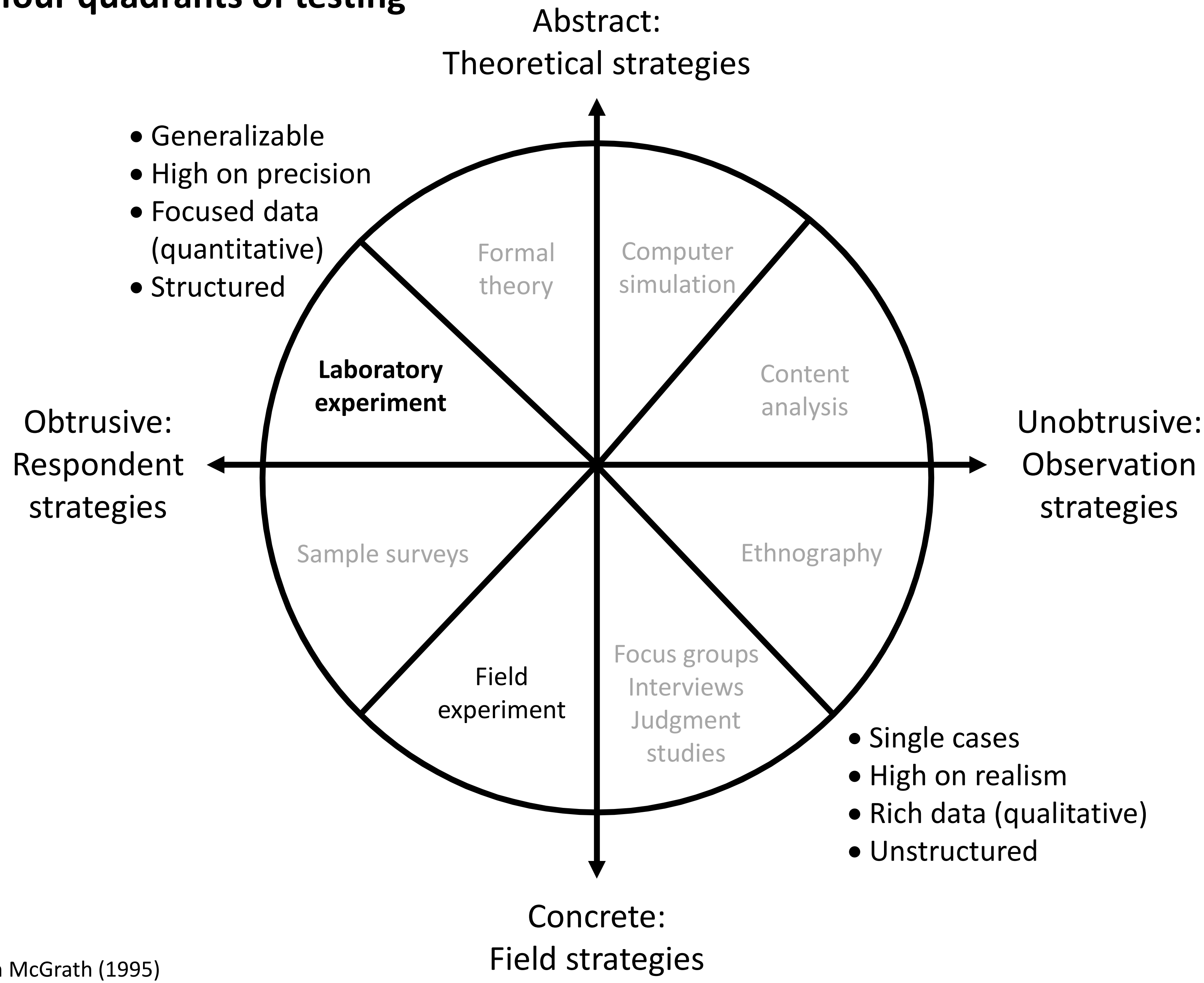
- ❑ Usability tests (formative-summative)
- ❑ Basic concepts
  - Research Question (RQ)
  - Hypothesis
  - Qualitative data
  - Quantitative data
- ❑ Data collection methods
  - Observation, interview, focus group
  - Lab and field experiments
- ❑ Measurement
- ❑ Data analysis
  - Exploratory
  - Confirmatory
- ❑ Research-informed System Design

Slides in this lecture are partially based on those from Ms. Tiffany Tang, Dr. Vincent Ng, Dr. Grace Ngai, Dr. Saul Greenburg, Dr. Christian Burgers, and Prof. Alan Dix et al.

What we are learning these two weeks

- ❑ Usability testing
- ❑ Experimental design in HCI
- ❑ Plan and set up an experiment
- ❑ Interpret results
- ❑ Statistics in HCI

The four quadrants of testing



Retake on McGrath (1995)

Laboratory experiments

- Definition of an experiment:
  - An experiment is designed to test a hypothesis about the role of one variable (the independent variable) on another (the dependent variable)

- In Confirmatory Data Analysis, or Statistical Hypothesis Testing, we have:
  - A hypothesis that we wish to “confirm” or “reject”
  - Two sets of measurements of the dependent variable, each corresponding to one configuration of the independent variables
  - A process that we will repeat with multiple subjects
  - A statistical method that we will use to compare multiple sets of data

Laboratory experiments

- The process of an experiment:
  - Variables that are not of interest are held constant while independent variables are manipulated
  - The effects of the manipulation on the dependent variables are observed

- To run an experiment, we need:
  - A hypothesis: Gamers’ involvement with Basim is higher than with Mario
  - Independent Variables (IV): the type of game character
  - Dependent Variable (DV): level of involvement rated from 0 to 5
  - Task: Play one mission in AC with Basim, play one race of 3 rounds with Mario
  - Analysis: Use statistical significance measures to compare the rate of involvement with Basim with the involvement rated for Mario

Mario. Mario Kart 8, Nintendo (2014)

Basim. Assassin’s Creed Mirage, Ubisoft (2023)

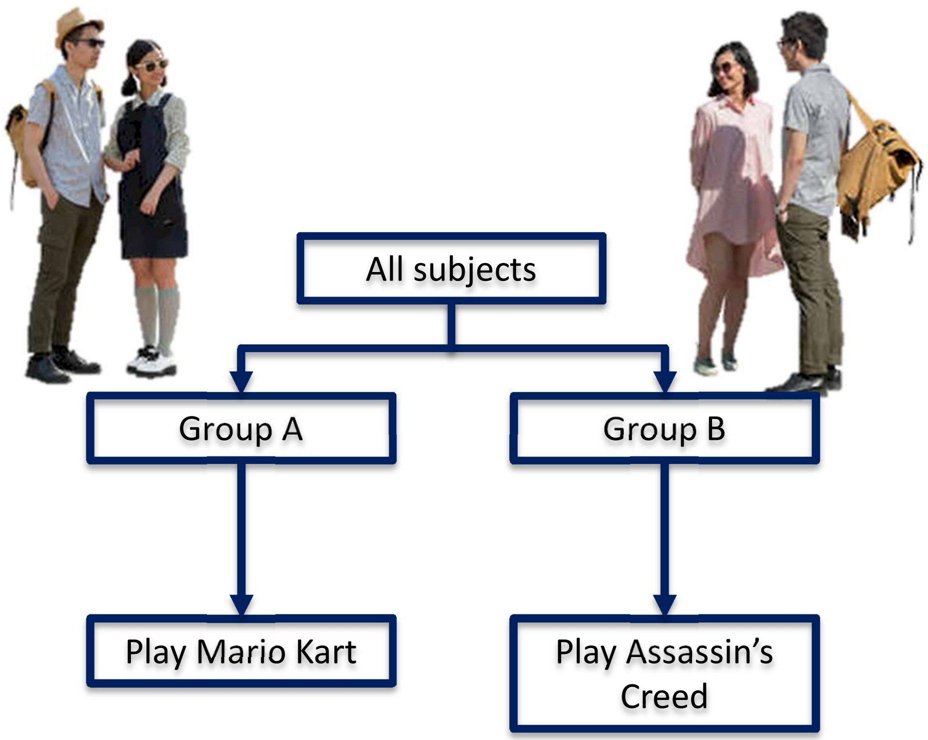
Experimental design

- Between-subjects
  - Split participants into two or more groups
  - Each group receives a different treatment / works under a different condition
  - Makes it easier to keep manipulations ‘clean’ (fewer order effects)
  - Variance is larger (different people, not just different conditions)

- Within-subjects
  - All participants receive all treatments / work under all conditions
  - Good when it is hard to recruit enough participants
  - Inevitable if you measure more dimensions
    (e.g., feeling happy or sad; level of usability and user satisfaction)
  - Variance is better comparable (same people under different conditions)

- Mixed designs
  - Between-subjects for the independent variable: Mario vs AC
  - Within-subjects for dependent measures: feeling involved, feeling distant

Experimental design



Between-subjects design

Beware of order effects!!

Within-subjects design

Between-subjects design

- Participants (users) split into two or more groups
  - Make sure the groups are demographically similar
  - Have equal amounts of training, experience, etc.

Mario players	Rating	AC players	Rating
Tom	0.5	Cindy	3.2
Dick	4.6	Joanna	5.0
Harry	2.1	Candy	2.1
Mary	3.5	Peter	4.3
Jane	4.7	Roy	2.5
Ida	5.0	Raymond	3.4
Carol	2.2	Victor	4.1
Nicholas	1.6	Jennifer	5.0
Tommy	4.7	Joyce	0.6
Jeanne	3.2	Jeremy	0.2
Florence	4.8	Ivan	4.9

Within-subjects design

- ❑ Same participants in all conditions: All participants perform the same task
  - Beware or order: - play Mario first, AC second
    - play AC first, Mario second

Players	Rating Mario	Rating AC
Tom	0.5	3.2
Dick	4.6	5.0
Harry	2.1	2.1
Mary	3.5	4.3
Jane	4.7	2.5
Ida	5.0	3.4
Carol	2.2	4.1
Nicholas	1.6	5.0
Tommy	4.7	0.6
Jeanne	3.2	0.2
Florence	4.8	4.9

Things to watch out for in experiment design

- The experimenter must systematically manipulate one or more independent variables in the domain under investigation
  - Mario vs AC; iPhone vs Blackberry; integrated vs dedicated Graphics Cards
- The manipulation must be made under controlled conditions, such that all variables that could affect the outcome of the experiment are controlled
  - Demographics: male – female balanced, similar age, education, cultural background
  - Experience: novice – expert
  - Habits: how much sleep, stress levels, how much coffee (Fitts!)
  - Computer performance at what temperature
- What might end up affecting the outcome of the experiment (aside from the independent variables?)
  - Mario is cartoonish and fun-oriented, AC is more realistic and aggressive
  - Task: Racing a kart is different from completing a kill mission
  - We measured feeling involved, how about feeling distant?

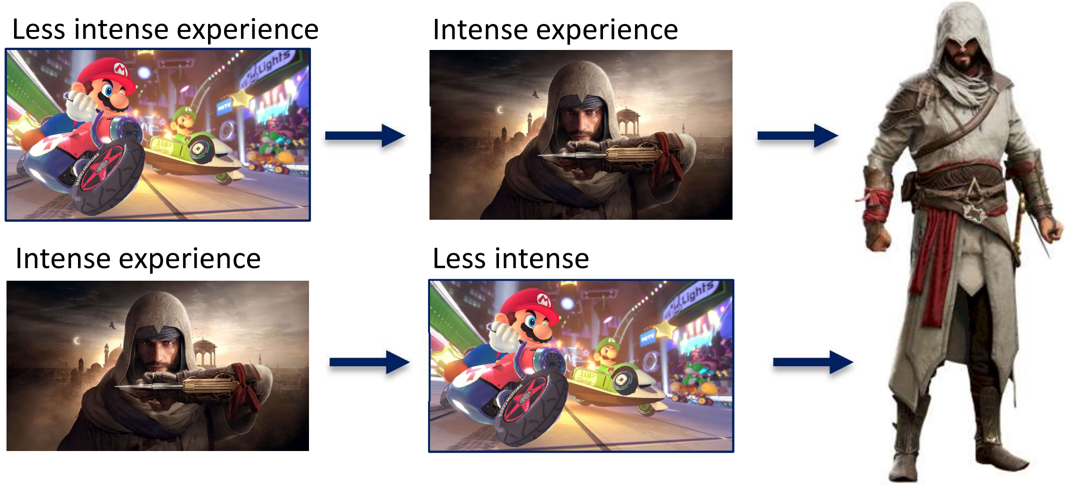
Confounding and order effects

❑ Confounds

• Happen when variables that were thought to be independent actually vary according to some other factor in the experiment (e.g., attitude, profession, or a novelty effect)

❑ Order effects

• A particular type of confounding, which involves the order in which the experiment tasks are carried out



The interaction with Basim Ibn Ishaq from Assassin's Creed was exciting

completely disagree	disagree	disagree a little	agree a little	agree	completely agree
0	1	2	3	4	5 ✓

Between-subjects but just for control; here, not of theoretical interest

The interaction with Basim Ibn Ishaq from Assassin's Creed was exciting

completely disagree	disagree	disagree a little	agree a little	agree	completely agree
0	1 ✓	2	3	4	5

Confounding effects

- ❑ Experience factors

- People have more/less relevant experience with one condition than the other
  Example: Foreigners do not use Alipay, Chinese do not use VISA much

- People in one group have more/less experience than the other group
  Usually happens when group division is not done properly in between-subject tests
  Example: Computing masters work with robot, Design bachelors with avatar
  (is the effect caused by the agent system or by education type/level?)

- ❑ Experimenter/subject bias

- Experimenter subconsciously treats subjects differently, or when subjects have different motivation levels (e.g., part of your sample are friends and family)
  Example: During instruction, experimenter treats boys different from girls;
  is enthusiastic about VR but not about the TV condition

- ❑ Uncontrolled factors

- Time of day, system load, but most notoriously: convenience sampling online!!!

Confounding effects

❑ Uncontrolled factors

• Convenience sampling online!!!

– Experimenter is lazy but wants large numbers of participants
– Invites friends and family, posts QR codes on Xiaohongshu
– Has no clue how surveys were filled out (or by whom)

Cat walks over keyboard, makes the choice for you

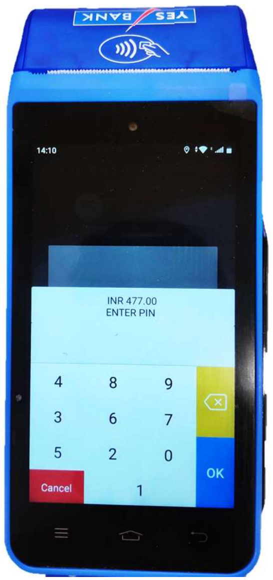
While taking survey in a noisy café, let’s make a selfie first

Order effects: within-subjects

- Tasks that are done earlier are slower and more prone to error;
  tasks that are done later are more prone to fatigue and boredom
- Warming-up effects (first condition trains for the second)

Payment terminals: Fixed layout vs shuffling input numbers for security reasons
Task demands on the user:
– Cannot rely on routine (= automated motor output)
– Has to pay more attention selecting proper keys
– More effort goes into memory retrieval

Fixed layout



Changing layout

Slowest and most prone to error, least fatigue and boredom

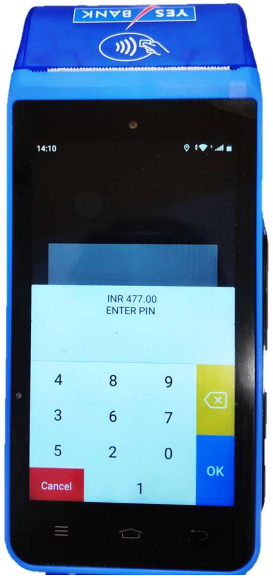
- Order effect of being 1st
- Novelty (not used to)
- Higher task demands

Fastest, least error, most prone to fatigue and boredom

- Order effect of being 2nd
- Overlearned
- Low task demands
- Sharp attention due to difficult first round

Moderate speed and error, moderate fatigue and boredom

- Order effect of being 1st
- Used to this most natural condition
- Low task demands



Moderate speed and error, moderate fatigue and boredom

- Order effect of being 2nd
- Warming up effect makes you faster
- Boredom compensated by higher task demands

One more confound: physical versus virtual keyboard!

Preventing confounding and order effects through randomization

❑ Control the effects of independent variable *X* on dependent variable *Y* so that effects are distributed randomly among conditions (groups)

- Randomly assign people to test Keypad Layout A first, or Layout B first
  (this addresses the order effect, hopefully it isn't significant)
- Randomly assign people to different groups
  (this addresses the experience effect, best would be equal distributions)
- All of this does not remove effects due to unknown differences,
  but ensures that any effect due to unknown differences among participants or conditions is random (and will be tackled through statistics)

Order effect

Playing Mario Kart was…	
fun	0 1 2 3 4 5
nice	0 1 2 3 4 5
happy	0 1 2 3 4 5
boring	0 1 2 3 4 5
tiring	0 1 2 3 4 5
sad	0 1 2 3 4 5

Mix!

Playing Mario Kart was…	
tiring	0 1 2 3 4 5
nice	0 1 2 3 4 5
boring	0 1 2 3 4 5
sad	0 1 2 3 4 5
fun	0 1 2 3 4 5
happy	0 1 2 3 4 5

Different orders for each participant can be accounted for as individual variance!

Preventing confounding and order effects through counter-balancing

❑ Counter-balancing

- Mitigates order effects in within-subject experiments

  Independent variables are varied across subjects in different orders

  Group 1: Tries Keypad Layout A first, and then Layout B
  Group 2: Tries Keypad Layout B first, and then Layout A

  If there is an order effect, this distributes the effect across all groups, but does not remove them

  This will fail, however, if order effects are not equal between conditions:
  Keypad Layout A is more effective at “warming up” people than Layout B…

  Therefore, four between-subjects groups with one within factor each: order

Group A

1^{st} 2^{nd}

within

Group B

1^{st} 2^{nd}

within

Group C

1^{st} 2^{nd}

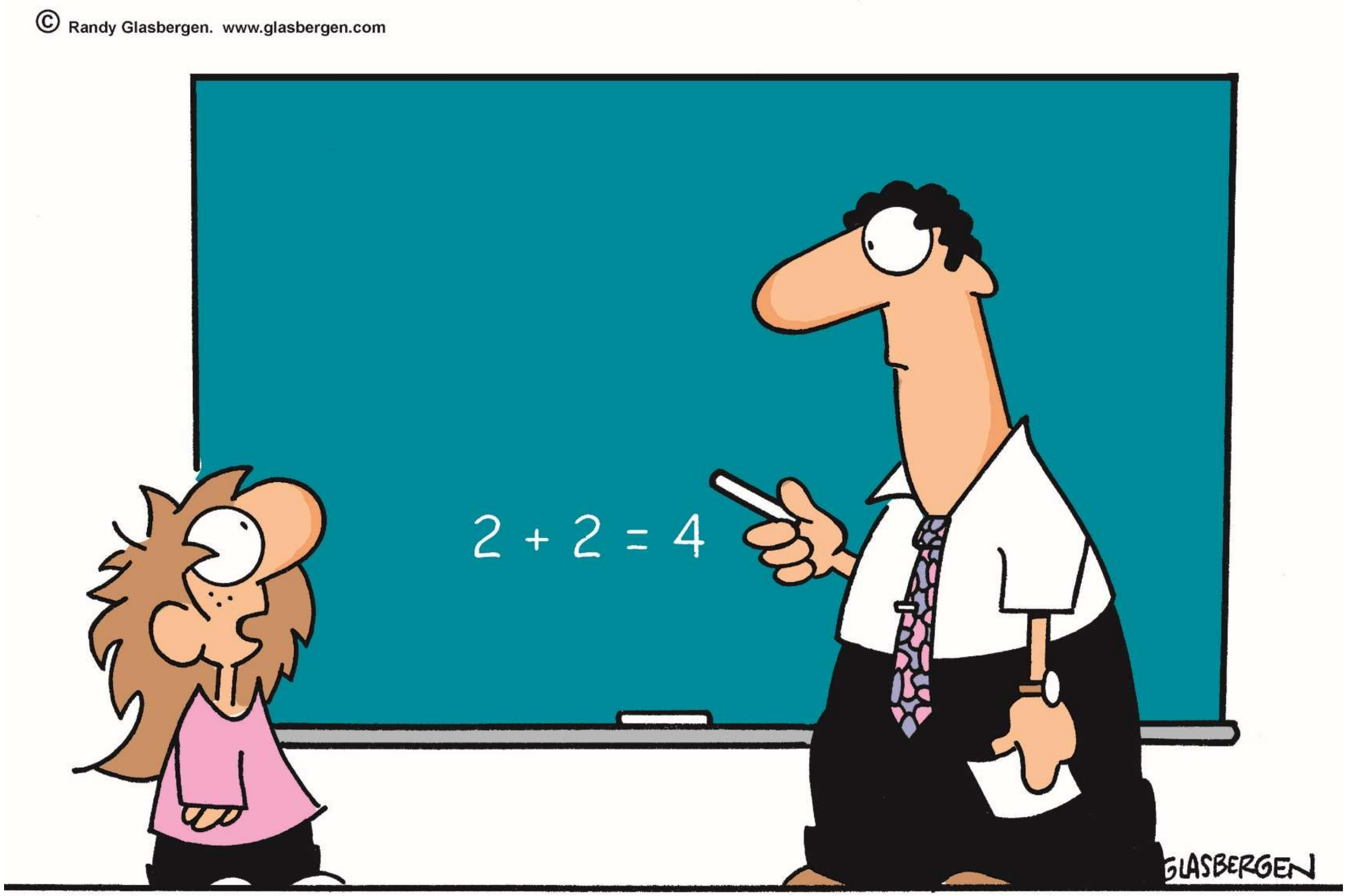
within

Group D

1^{st} 2^{nd}

within

Next: statistical analysis



"How can I trust your information when you're using such outdated technology?"

Statistical analysis

- ❑ Calculations that tell us mathematical attributes about our data sets
  - Mean, amount of variance, ...
- ❑ How data sets relate to one another
  - Whether we are “sampling” from the same or different distributions
- ❑ The probability that our claims are correct
  - “statistical significance”

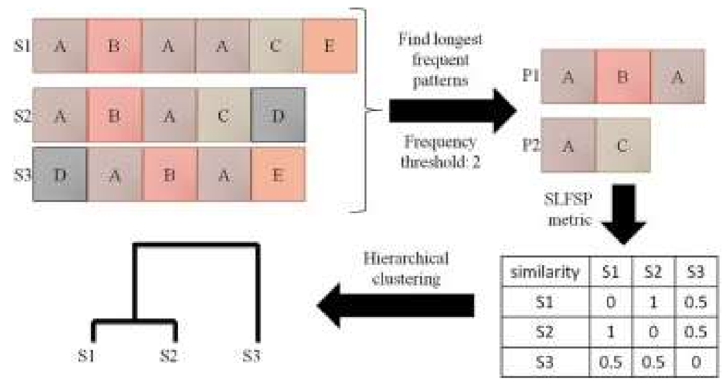
Exploratory and confirmatory data analysis

- ❑ Exploratory data analysis enables you to find out potential patterns from collected data
- ❑ Confirmatory data analysis allows you to validate whether the collected pattern really holds (normally)

- Without any pattern discovered, there is nothing for validation!
- Nevertheless, validation result can provide feedback for further exploratory study

```
input : S: A set of activity sequences. min_fr: A frequency threshold.
output: allLongestFrequentPatterns: A set of longest frequent patterns.
begin
    frequentActivities ← Get Vocabulary Activities(S, min_fr);
      /* Algorithm 2 */
    k = 2;                                  /* Length of patterns */
    frequentPatterns_{k-1} ←frequentActivities;
    while |frequentPatterns_{k-1} | > 0 do
        candidatePatterns ← Get Candidate
          Patterns(frequentPatterns_{k-1}, frequentActivities, k);
          /* Algorithm 3 */
        frequentPatterns_k,allLongestFrequentPatterns ← Get Frequent
          Patterns(min_fr, k, S, candidatePatterns);   /* Algorithm 4 */
        k++;
        frequentPatterns_{k-1} ←frequentPatterns_k;
    end
    return allLongestFrequentPatterns;
end
```

Distinguishing surgical behavior by sequential pattern discovery

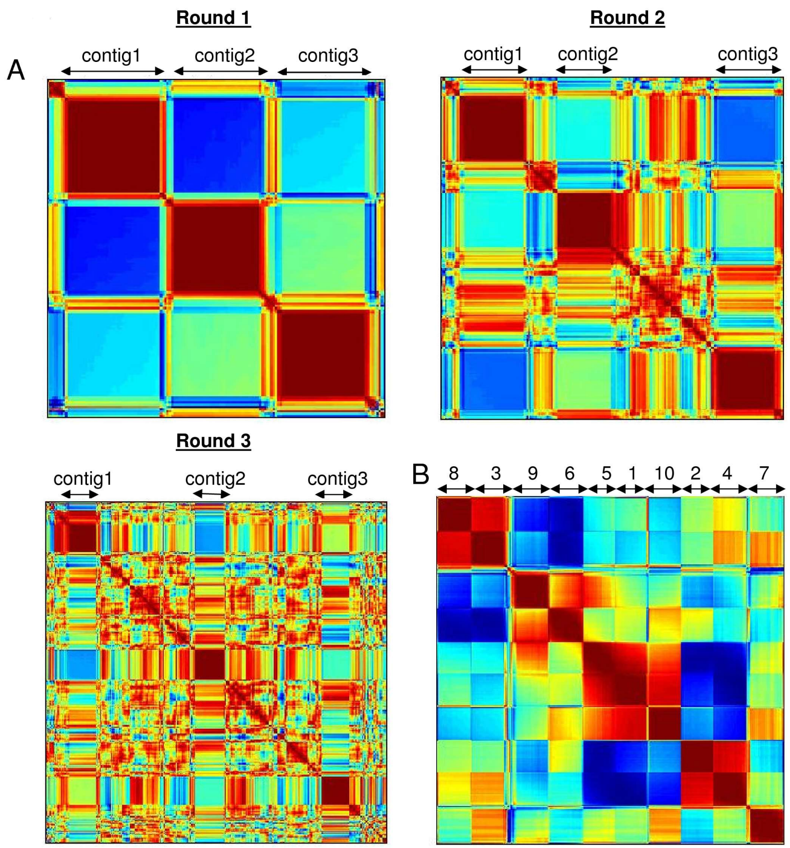


Data mining for pattern discovery: exploratory or confirmatory?

Huaulmé, A., Voros, S., Riffaud, L., Forestier, G., Moreau-Gaudry, A., & Jannin, P. (2017). Distinguishing surgical behavior by sequential pattern discovery. *Journal of Biomedical Informatics*, 67, 34-41.

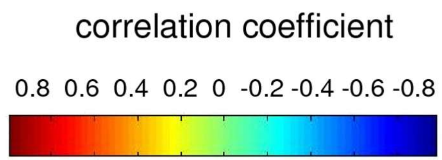
Exploratory data analysis

- Inductive approach, “descriptive statistics”
- Analyses data sets to summarize main characteristics, looking for patterns
- Often uses visual methods to aid analysis

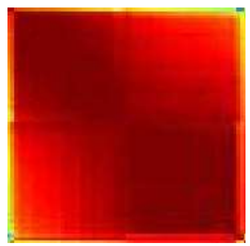


Visualization-driven discovery of co-occurrences in gene-expression

Red is high positive correlation, interpreted as higher similarity in gene expression



Particularly pattern 5 and 1 in the middle of B



Morozova, O., Morozov, V., Hoffman, B. G., Helgason, C. D., & Marra, M. A. (2008). A seriation approach for visualization-driven discovery of co-expression patterns in Serial Analysis of Gene Expression (SAGE) data. *PloS one*, *3*(9), e3205. doi: 10.1371/journal.pone.0003205.g001

Exploratory data analysis

- ❑ Inductive approach, “descriptive statistics”
- ❑ Analyses data sets to summarize main characteristics, looking for patterns
- ❑ Often uses visual methods to aid analysis

Lecture on Visuals:

Pattern recognition

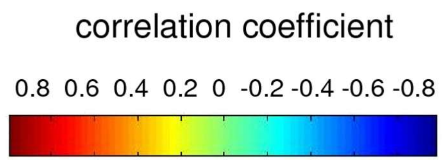
Pattern 5 and 1 are placed into close *proximity* based on __________?

similarity!

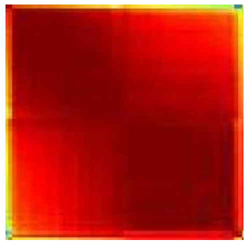
• Where is the theory that says that high correlation indicates similarity?
• Where is the theory that says that things highly similar should be placed together?

Visualization-driven discovery of co-occurrences in gene-expression

Red is high positive correlation, interpreted as higher similarity in gene expression



Particularly pattern 5 and 1 in the middle of B

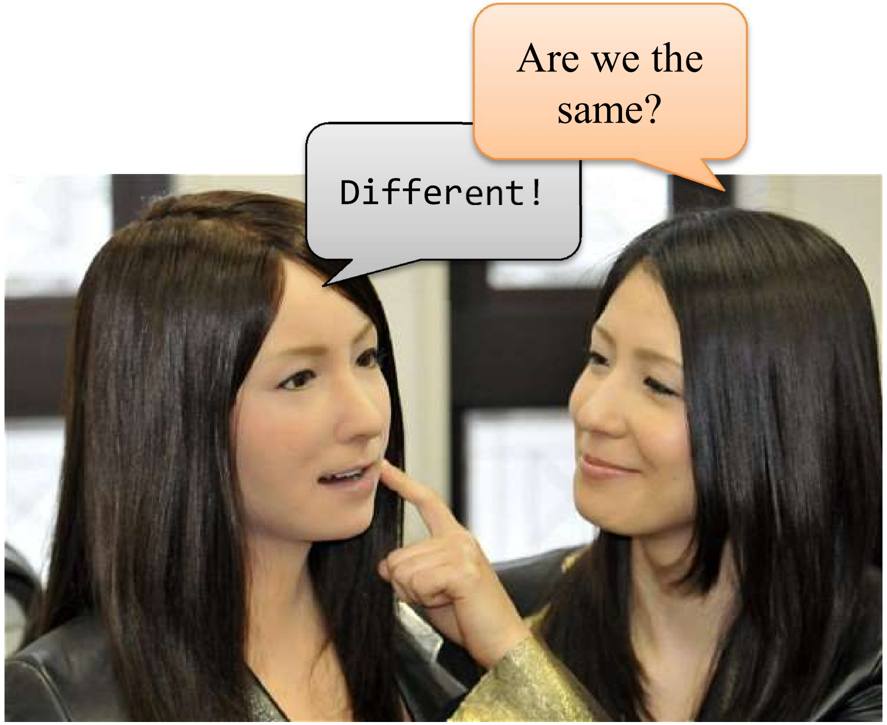


Exploratory data analysis

- Good to come up with hypotheses that then later can be / should be tested (provides a hypothesis, not a conclusion)

  - Let’s say we want to know the relation between similarity and liking

    – More similar is more liking?
    – Is the increase linear? Log-linear? Exponential?
    – Is there an optimum? Parabolic? Cubic?

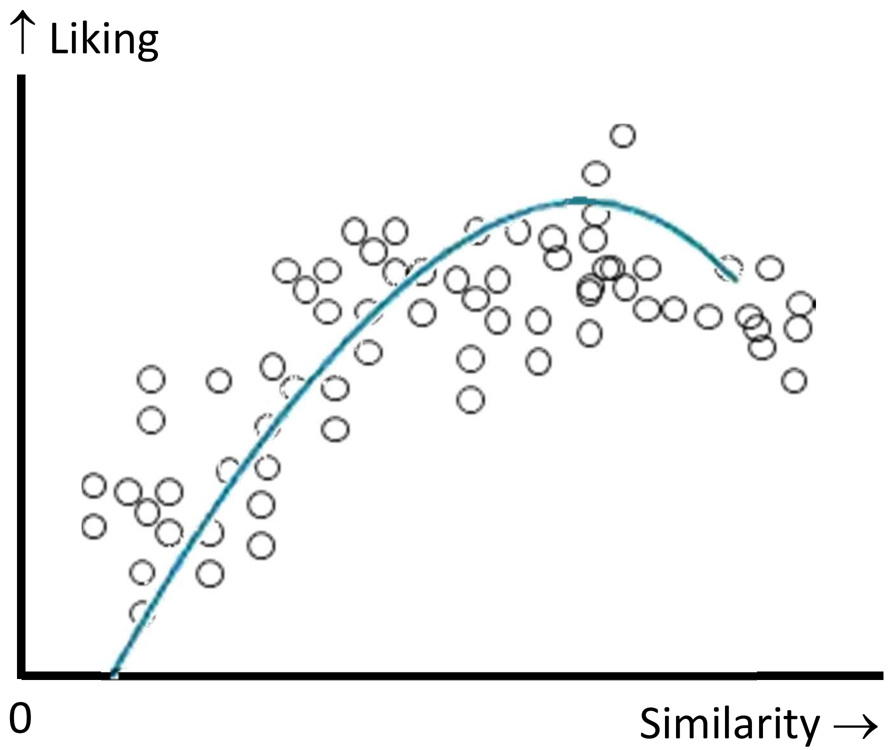


So we run a questionnaire:

Robot and I look alike	0 - 5
Robot and I resemble each other	0 - 5
Robot and I are the same	0 - 5
I like the robot	0 - 5
I appreciate the robot	0 - 5
I favor this robot	0 - 5

Exploratory data analysis

- To find the relation between similarity and liking, plot the data points
- Do curve fitting, using SPSS: Analyze > Regression > Curve Estimation
- Excel: X-Y scatter plot, right click curve,
  Add trend line, Polynomial, Order 2
  Then, Options > Display equation on chart



– It's a parabola
– We found an optimum hypothesis
– The equation is: $-\frac{1}{2}x^2 + 4x - 3$

Now we should run an experiment with the hypothesis H_1:

There is an optimum to liking (y) a robot based on its similarity (x) to its user, according to $y = -\frac{1}{2}x^2 + 4x - 3$

and do Confirmatory data analysis

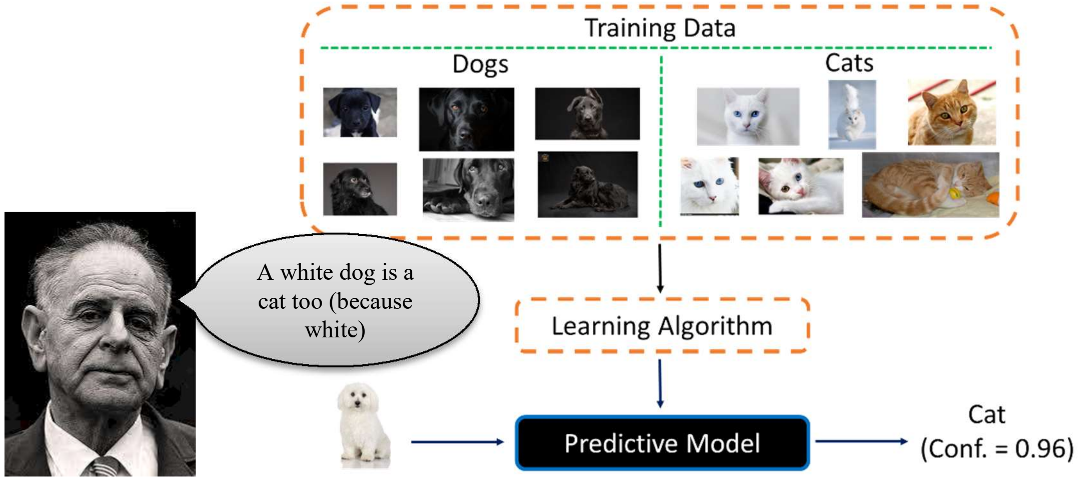
Confirmatory data analysis

- ❑ Deductive approach — inferential statistics
- ❑ Looks for definite answers to specific questions
- ❑ Often uses probability models and numerical calculations
- ❑ Given a hypothesis determined at the outset, do the data support our hypothesis?

- Confirmatory data analysis uses Descriptives to **summarize** and **describe a group of numbers** from a research study, but that's just the start
- Inferential statistics are used to draw conclusions and to make inferences that go beyond the individual cases
- Allows researchers to make inferences about a large group of individuals (i.e. population) based on a research study in which a much smaller number of individuals (i.e., sample) took part

Confirmatory data analysis

- ❑ Testing the $H_0 \rightarrow$ Can 'no notable difference' be refuted?
  - Statistically, we can never demonstrate that anything is 'correct' or 'true,' only that it is *very likely* to be correct: "Most swans are white"
  - Statistics can also underscore that something is *very likely to be false*, finding a number of swans that are not white
  - The stronger the statement, the less sophisticated statistics I need: "All cats are white" and "Anything white is a cat" are easily falsified by finding one counter-example (cf. Karl Popper)
- ❑ The Null is conservative and only rejected if it is significantly *unlikely* that no-difference / no-effect occurred



Lakkaraju, H., Kamar, E., Caruana, R., & Horvitz, E. (2017). Identifying unknown unknowns in the open world: Representations and policies for guided exploration. In S. Singh & S. Markovitch (Eds.), *Proceedings of the 31st AAAI Conference on Artificial Intelligence, Feb. 4-7, 2017, SF, CA* (pp. 2124–2132). Palo Alto, CA: AAAI.

Central tendencies

- The central tendency of a group of scores (a distribution) refers to the middle of the group of scores

- Mean, written as M, μ, or $\bar{X}$ (X-bar)
  - X stands for the scores in the distribution of the variable X
  - N or n stands for the number of scores in a distribution

Sample Mean	Population Mean
$\bar{x} = \frac{\Sigma x}{n}$	$\mu = \frac{\Sigma x}{N}$

where Σx is sum of all data values

N is number of data items in population

n is number of data items in sample

- Mode
  - The mode is the most common single value in a distribution

- Median
  - If you line up all the scores from lowest to highest, the middle score is the median
  - If there are two middle scores, the median is the average of the two

Visual inspection of central tendencies is not reliable

- Given any set of collected data, there will almost always be some variation

  - Differences between data sets may be due to just normal variation!
  - Two sets of ten tosses with different but fair dice

  – First dice gives 6, 2, 4, 4, 2, 6, 4, 2, 5, 6 (M = 4.1)
  – Second dice gives 6, 4, 4, 5, 4, 4, 2, 1, 4, 5 (M = 3.9)

If they are both fair dice, shouldn't their outputs have the same probability? Does this mean that the dice are not fair? Given any set of data (even dice throws), there will *always* be some variation

- The question is whether this variation is explainable by normal variation (Do you believe that the world is fixed and stable or that everything moves and changes?)

- Is a difference of 0.2 explainable? Is the average stable enough? What about 1000 attempts?

Variance

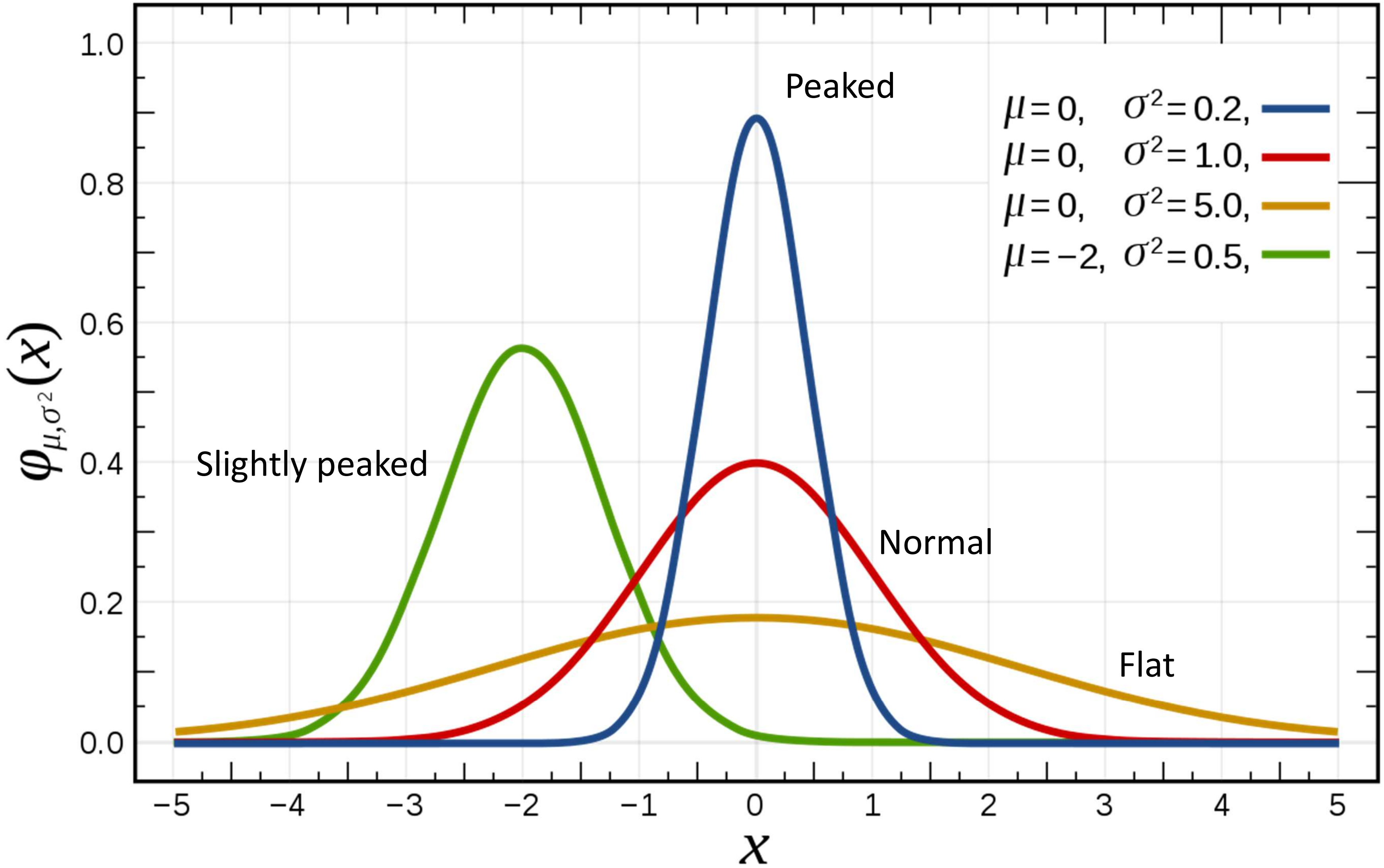
- How spread out are the scores in a distribution?
  - X stands for the scores in the distribution of the variable X
  - M, μ, or $\bar{X}$ is the mean
  - N or n stands for the number of scores in a distribution

- Standard deviation (σ, S, or *SD*)
  - The square root of variance

	Population	**Sample**
Variance	$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$	$S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{n-1}$
Standard deviation	$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$	$S = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{n-1}}$

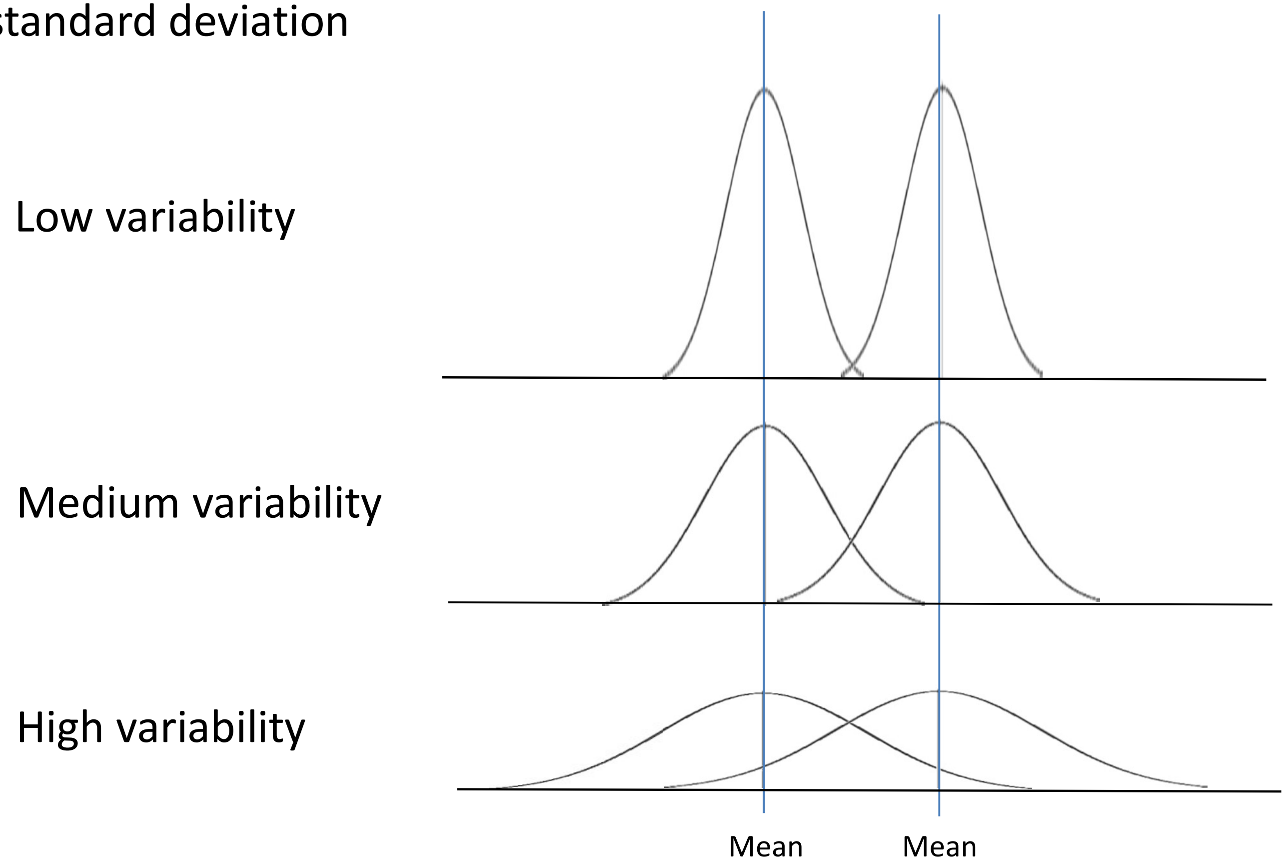
The divisor $n - 1$ makes the result slightly larger, compensating for the underestimation by the sample mean of the population *SD*

Variance, some examples for different distributions



Mean and variance

- There are two separate issues here: the mean, or the average, and the variability, or the standard deviation



- The difference between the means of the two curves is the same in all three above situations. However, they certainly do not look alike!

Signal versus noise

- Given two datasets, the difference between their means is the "signal"
  - The change that we think that we introduced into the dataset by manipulating the independent variables
- The variability is the "noise" that is introduced by random factors beyond our control
  - Given any dataset, there will always be some "noise," so we cannot trust that the "signal" tells the story 100%
- The question is: How much noise can we expect to encounter in a normal situation?
  - Meaning: When can we trust that the difference that we're seeing is a reliable difference, and not just because of noise?

Signal = mean, noise = variance

- Intuitively, we can trust that the difference we are seeing really is a difference when:

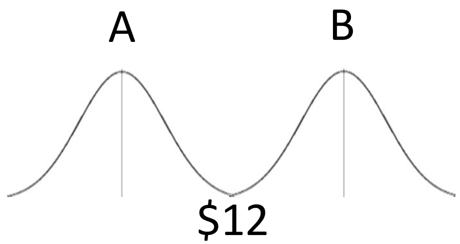
  - We have more observations in our sample

    – A political poll of 10,000 people that gives Candidate A an average rating of 36 is more believable than a poll of only 100 people

  - The datasets themselves have smaller variability

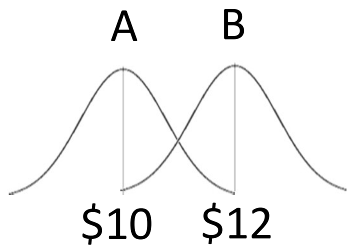
Hong Kong Island
Suppose prices in Shop A vary between \$10-\$12
and in Shop B between \$12-\$14

Prices don't overlap, Shop A is cheaper than Shop B



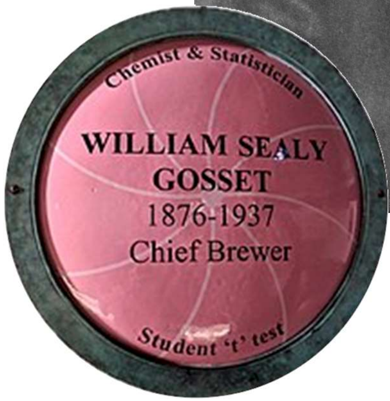
Sham Shui Po
Shop A's prices are between \$ 8-\$12
and Shop B's are between \$10-\$14

Harder to tell whether Shop A is cheaper or not



Student's *t* and signal to noise ratio

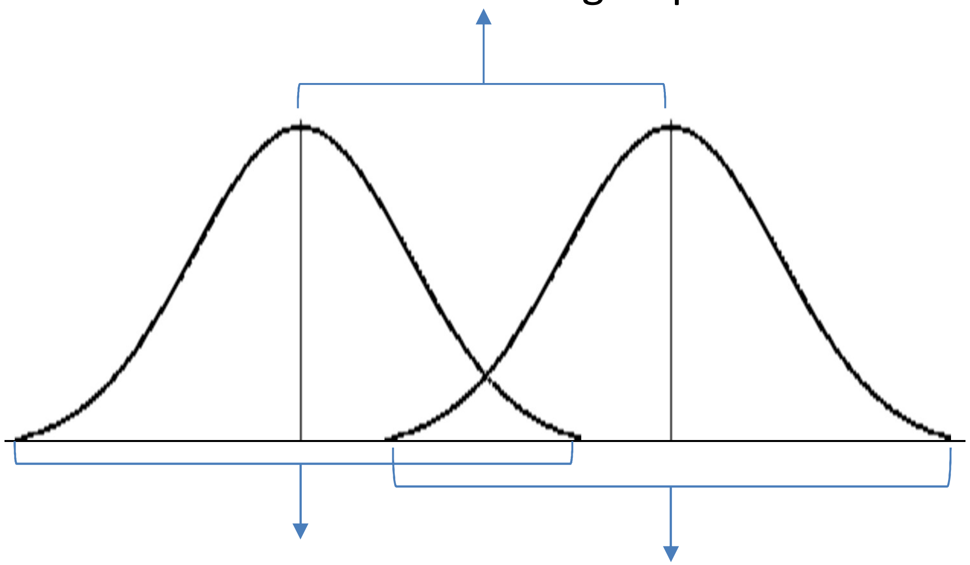
- Developed by William Sealy Gosset over 100 years ago!
    - The *t*-test was originally used by Gosset to compare differences in barley yields (for a beer brewery)
- The idea is to calculate the signal (mean difference) to noise (variability) ratio
- A large part of Western-based science is indebted to the *t*-test and related techniques such as (M)ANOVA



T-test (independent samples)

- A statistical test that compares the difference between the means of two groups of data, taking the variability of the two groups into account
- Calculates a *t*-value which gives us the signal-noise ratio
- If this ratio is high enough (we have enough "signal" to compensate for the "noise"), then we can be more confident that the signal is trustworthy

Signal Difference between group means $\bar{x}_1 - \bar{x}_2$



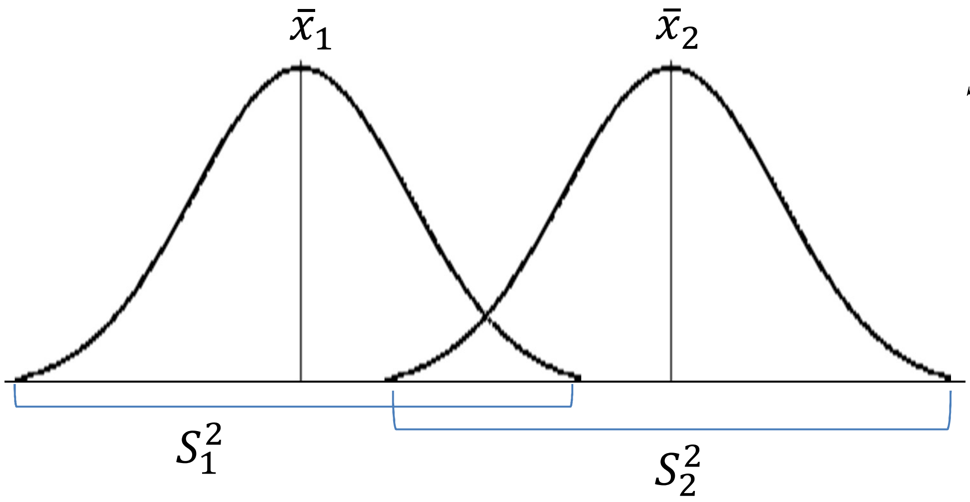
Noise Variability of groups $\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$

T-test (independent samples)

- A statistical test that compares the difference between the means of two groups of data, taking the variability of the two groups into account
- Calculates a *t*-value which gives us the signal-noise ratio
- If this ratio is high enough (we have enough "signal" to compensate for the "noise"), then we can be more confident that the signal is trustworthy

Numerator
Denominator

$$\frac{\textbf{Signal}}{\textbf{Noise}} = \frac{\text{Difference between group means}}{\text{Variability of groups}} = t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$



S is the standard deviation, so S^2 is the variance

T-test: different types

- Independent samples *t*-test (a.k.a., two-samples *t*-test)

  - Compares the difference between the means of **two groups** of data, taking the variability of the two groups into account

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

This is the equation we just discussed

- One-sample *t*-test

  - Tests the mean of **a single group against a known value**

- Paired samples *t*-test

  - Compares means from the **same group** at different times (e.g., before-after treatment) or different conditions in a within-subjects experiment (e.g., working with an avatar vs a robot, order effects, or measuring happy versus sad)

T-test: one sample

- One-sample *t*-test

- Tests the mean of **a single group against a known value**

μ_0 is the known value

S/sqrt(n) is the Standard Error of the Mean (SEM), indicating how accurately your sample estimates the mean of the population

$$t = \frac{\bar{x} - u_0}{s/\sqrt{n}}$$

$$SEM = \frac{s}{\sqrt{n}}$$

Example: Does the mean IQ score of COMP students significantly differ from the population average (which is 98, the known value)

(The answer is ‘yes’)

T-test: paired samples

- Paired-samples t-test compares means from **the same group** at different times (e.g., one year apart) or different within-experiment conditions (e.g., using both iPhone and Blackberry)
  - Interestingly, paired samples t-test and one sample t-test are mathematically equivalent
    - Known mean is 0
    - Single group is now the group of the differences

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{\bar{x}}} \longrightarrow t = \frac{\boxed{(\bar{x}_1 - \bar{x}_2)} - 0}{s_{\bar{x}}}$$

Estimated Standard Error of the Mean

SEM of the new group $\longrightarrow$ $s_{\bar{x}} = \frac{s_{diff}}{\sqrt{n}}$

The sample standard deviation of the differences

The role of the normal distribution in a *t*-test

- The normal distribution allows the *t*-test to say something about how likely it is that the differences between the means of the two groups are systematic, or just happened by matter of chance

  - Default is the Null (H_0) = observed differences are chance occurrences
  - Given enough data, or a small enough variability, we can reject the Null in favor of the Alternative (H_A or H_1) if:

    – There is probably a systematic difference between the means, and
    – I am so-and-so much confident in my decision (e.g., 95% convinced)

  - If the *t*-test shows that the difference between the means (the signal) is more than what would be expected purely by chance,

    – then we say that the difference is **statistically significant**
    – Certain researchers would then **reject** H_0 and **accept** H_1

> Be aware that the 'frequentist approach' illustrated here can only reject the Null or fail to do so; it cannot demonstrate the Alternative. Moreover, detecting no notable difference does not say that there is no difference. It also does not say that the two samples were drawn from the same population!

Significance: degrees of freedom and rejection area

- If the *t*-test shows that the difference between the means (the signal) is more than what would be expected purely by chance, then we say that the difference is statistically significant. How does the *t*-test "show" us this with a value?

  - The numerator of the *t*-value is the difference between the means of the groups
  - The denominator is the variability of the groups

- Intuitively, the larger the *t*-value is, the more significant the difference (or the signal) is, compared to the noise

  - Significance depends on the size of the dataset (the degrees of freedom) (i.e. the number of scores that are free to vary, see next slide)
  - And on the risk level (or alpha level or rejection area) that we set: $\alpha = .05$

  – Gives the level of confidence we want to have before making a judgement
  – Usually set at 95% (or a 5% chance that we might be wrong): $p < .05$
  – But that is pretty arbitrary and depends on the situation
  (10% patients cured may not be 'significant' but may save thousands of lives)

Significance: degrees of freedom (df)

❑ Bonus points for doing more observations

- Degrees of freedom are the maximum number of values that you can freely vary in a data sample while keeping the same mean
- Calculated as the sample size minus the number of restrictions

– One-sample t-test: $df = n - 1$
– Independent samples t-test: $df = n_1 + n_2 - 2$

Suppose I have three scores with Mean = 5. Two of the values are 4 and 6, so the third value has to be 5. The scores 4 and 6 could also be 3 and 7 but I need to fix 5 or else Mean ≠ 5. With 3 scores of which 2 are free to vary, df = 3 – 1 = 2 (so if you only have one sample, $df = n - 1$)

$N = 200$
Male, $n_1 = 98$
Female, $n_2 = 102$

Independent samples t-test: $df = n_1 + n_2 - 2$

Effect of Males versus Females, $t_{(df)} = t_{(198)} = 5.49$ or whatever…

Significance: degrees of freedom (df)

- Put more formally, if you know $\bar{X}$ with the sample size n and if you know the scores $X_1, X_2, \ldots, X_{n-1}$, then you can calculate the value of X_n

- In other words, $\bar{X}$ has $n - 1$ degrees of freedom

$$X_n = n\,(\bar{X}) - \sum_{i=1}^{n-1} X_i$$

- Level of significance, then, is the *t*-statistics *in relation to* the degrees of freedom

  • More degrees of freedom and the *t*-value can be lower to establish a significant effect
  • Fewer degrees of freedom and the *t*-value has to be very high to establish a significant effect

Table of significance for a two-tailed *t*-test

df
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

Here we have an array of degrees of freedom

How many people in this experiment?

2

How many people in this experiment?

21

Table of significance for a two-tailed *t*-test

	20% error	10% error	5% error	I accept only 2%	Chance is that I am 1% wrong
df	**0.2**	**0.1**	**0.05**	**0.02**	**0.01**
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					

These are levels of significance (the error we are willing to tolerate)

$\alpha = 0.20$ means I accept to be 20% of the times wrong

$\alpha = 0.05$ means I accept to be 5% of the times wrong

So the probability *p* that my effect is not systematic and unreliable should be smaller than 5%

So it must be that $p < \alpha = .05$ before I reject the H_0 (and accept the H_A)

Minimal *t*-values we need, given number of observations (df) and error margin (significance level)

df	0.2	0.1	0.05	0.02	0.01
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.92	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.44	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.86	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.25
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.35	1.771	2.16	2.65	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.12	2.583	2.921
17	1.333	1.74	2.11	2.567	2.898
18	1.33	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845

These *t*-values are the critical minimum signal-to-noise ratios

Table of significance for a two-tailed *t*-test

With more observations, *t*-value can be lower

df	0.2	0.1	0.05	0.02	0.01
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.92	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.44	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.86	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.25
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.35	1.771	2.16	2.65	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.12	2.583	2.921
17	1.333	1.74	2.11	2.567	2.898
18	1.33	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845

Table of significance for a two-tailed *t*-test

df	0.2	0.1	0.05	0.02	0.01
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.92	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.44	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.86	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.25
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.35	1.771	2.16	2.65	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.12	2.583	2.921
17	1.333	1.74	2.11	2.567	2.898
18	1.33	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845

With less acceptance of doubtful effects, *t*-value must go up

Probability: tolerance of error, level of confidence

- SPSS, R, or Excel (and other spreadsheets and statistical packages) calculate the probability = *p*-value for us

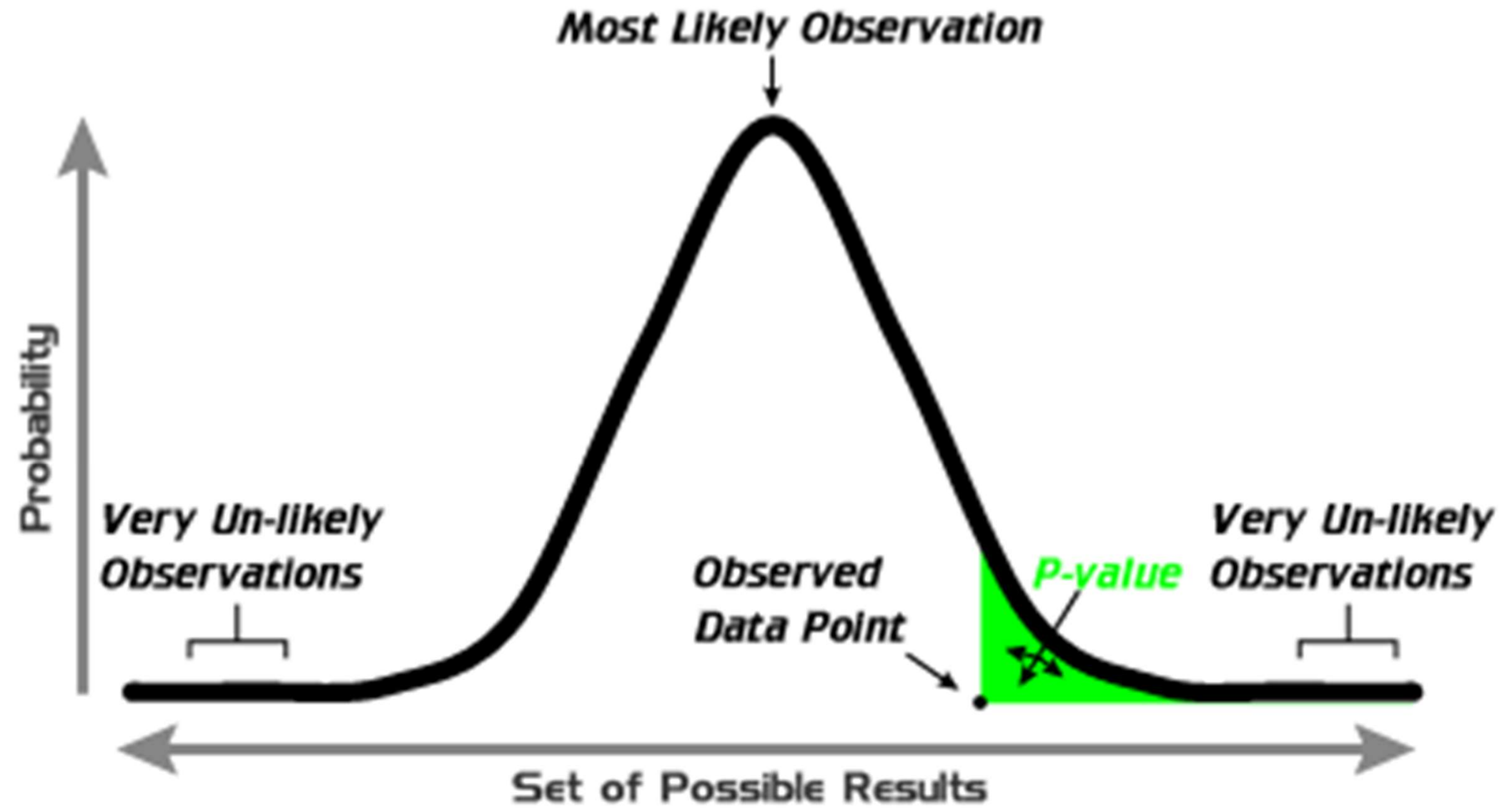
- The *p*-value gives us the confidence that the Null hypothesis is valid

    - For instance, a *p*-value of 0.30 means that we are 30% sure that there is no significant difference between the two datasets

- Therefore, we reject the Null hypothesis if the *p*-value is less than or equal to our required significance value -- i.e. the amount of error that we're willing to tolerate under the level of confidence that we require

    - For instance, if we are using a significance value of 0.05 (or we want a confidence of 95%), we reject the Null hypothesis if the *p*-value is 0.05 or less

Probability: tolerance of error, level of confidence



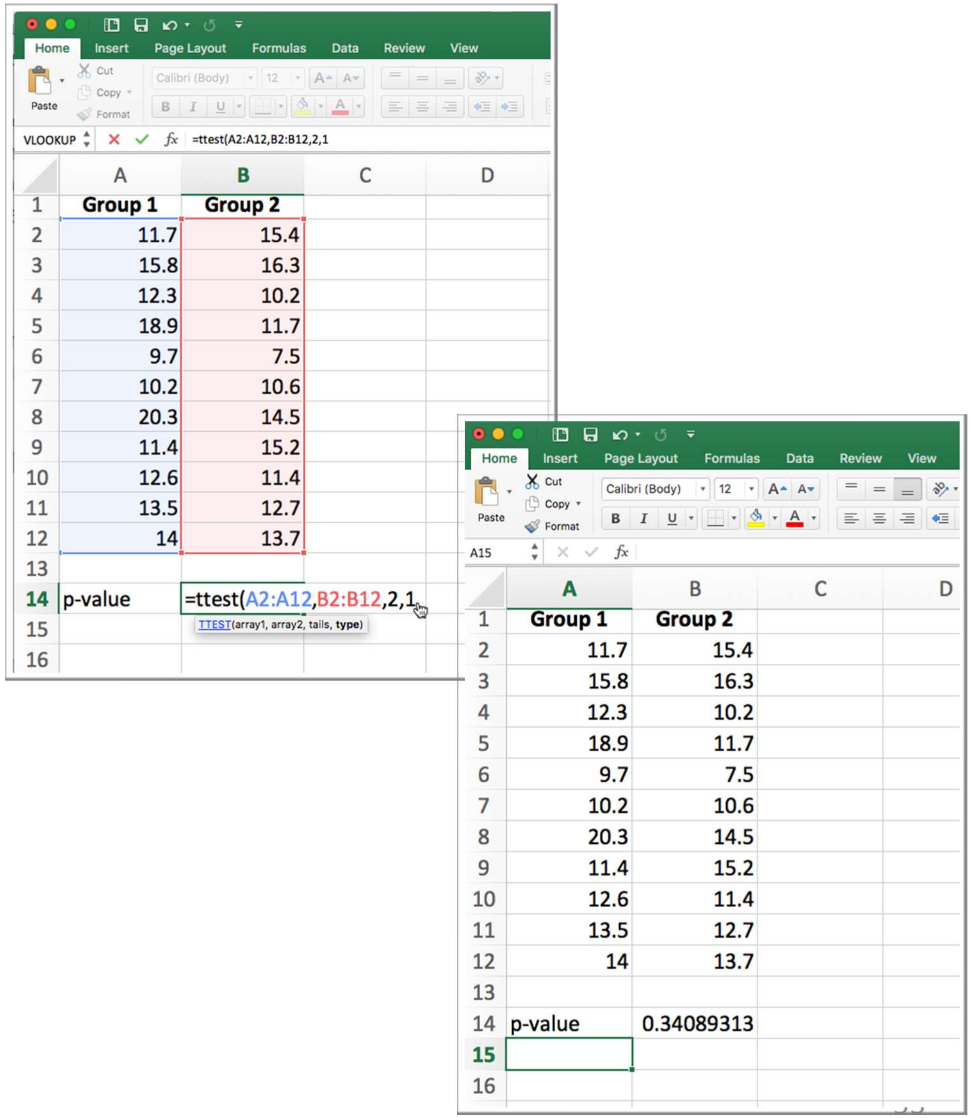
A p-value (shaded green area) is the probability of an observed (or more extreme) result arising by chance

T-test in Excell

ttest(array1, array2, tails, type)
array1: The first data set
array2: The second data set

tails: number of distribution tails
1: one-tailed
2: two-tailed

type : The kind of t-test to perform
1: paired test
2: two-sample equal variance
3: two-sample unequal variance



Some limitations of the *t*-test

The *t*-test is very widely used, but it does have some limitations:

- ❑ Data points of each sample are normally distributed
  - Not always the case, but *t*-test very robust in practice, due to Central Limit Theorem (CLT)

- ❑ Population variances assumed equal
  - *t*-test reasonably robust for differing variance, however, does deserve consideration, may use *F*-test to validate (ANOVA / MANOVA)

- ❑ Individual observations of data points in sample should be independent

- ❑ Significance level
  - Decide upon the level before you do the test!
  - Typically stated at the .05 or .01 level

CLT: A sample approximates normal distribution as the sample size gets larger (i.e. aim for $n \geq 50$ in each Between-Ss condition to avoid trouble)

False alarms (Type 1 error) and Missed signals (Type 2 error)

- The *t*-test gives us the probability that the difference that we're seeing between the data sets is not due to pure chance
  - What sort of significance value should we aim for? Is 95% too high?

- There are two types of mistakes we might make
  - Type 1 error: Null hypothesis is true, but we reject it by mistake (false alarm)
  - Type 2 error: Null hypothesis is false, but we accept it by mistake (missed signal)

- Effects of the level of significance

  one ten-thousandth
  - If we set our confidence level too high (e.g., $p < 0.0001$), we run a greater chance of Type 2 errors
  - If we set our confidence level too low (e.g., $p > 0.1$), then we run a greater chance of Type 1 errors

		Reality	
		Null hypothesis is true	Null hypothesis is false
Experiment	Null hypothesis is true	Accurate	Type 2 error
	Null hypothesis is false	Type 1 error	Accurate

missed signal!

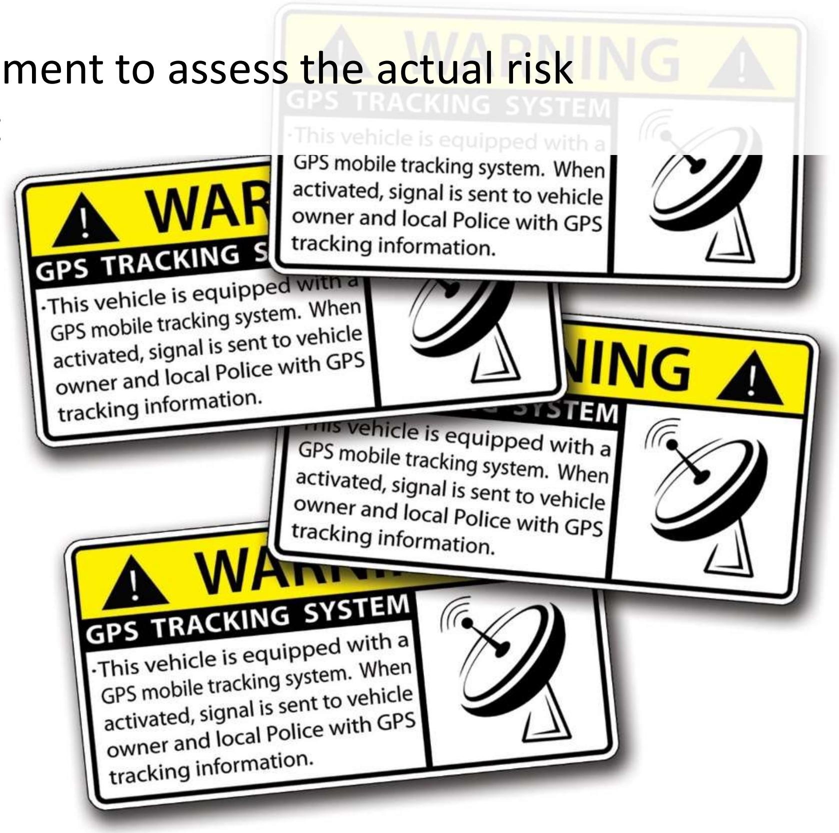
false alarm!

False alarms (Type 1 error) and Missed signals (Type 2 error)

- ❑ In most scientific studies, Type 1 errors are generally considered to be worse because the hypothesis is the current state-of-the-art theory. Therefore, people expect a high confidence level if you're going to overturn it
  - • For example, if it's "presumed innocent till proven guilty," you need an overwhelming vote in a jury to convict a serious crime (e.g., 7:2 or 8:1)

- ❑ However, you must use your judgement to assess the actual risk of being wrong, given your context

Maybe in threatening circumstances you should be cautious at the risk of calling out a false alarm (but may cause alert fatigue) →



False alarms (Type 1 error) and Missed signals (Type 2 error)

Suppose we are redesigning the Flight Management System (FMS) from Lecture 1, using Sparse Haptic Proxy from Lecture 5

- ❑ What is the consequence of making a Type 1 or 2 error, if we consider user performance in terms of accuracy?
  - H_0: Using the new touch panel does not result in more correct user responses
  - H_1: The new touch panel leads to more correct responses

False alarms (Type 1 error) and Missed signals (Type 2 error)

- A Type 1 error would mean that the new FMS does not give us better performance, but we change it anyway. Consequence: We spend extra resources developing the software, and our pilots must learn a new GUI for no benefit

- A Type 2 error would mean that the new FMS will give us a more accurate performance, but we stick with the old menu to “play safe” (upon pilot request). Consequence: The pilots make more errors typing in ROZO or ROMEO, crashing their planes into a mountain but the interface is already familiar to them

Testing in HCI may save lives

Human-generated summary

User testing is important to inform design decisions: Judgments by experts, lab and field experiments, and field studies with real users avoid market failure (irrespective of proper coding or design method)

Start testing while prototyping ('formative'). Don't wait until system roll out ('summative')

Qualitative tests generate non-numerical data (e.g., videos, use stories), which give you specific insights into your particular case

Quantitative tests are numerical (e.g., speed, accuracy, attitude) and may be generalized over multiple cases

Human-generated summary

Experiments are great for quantitative results. Yet, choose the right measures, know its scale (ratio, nominal, &c), avoid confounds or order effects by randomizing and counter-balancing

Analysis may be exploratory. Don’t jump to conclusions, you haven’t tested anything yet. You may now have a hypothesis that needs a controlled experiment to do confirmatory statistics on the new data

T-tests are simple techniques to inspect two sample means. Yet, experiments often are more complicated. Study General Linear Models: uni-variate, multivariate, and repeated measures. Get acquainted with Bayesian statistics if H_0 (no difference) is theoretically relevant (cf. human-level AI performance or robot portraits of users)

Human-generated summary

Controlled experiments can provide clear convincing results on specific issues

Creating testable hypotheses are critical to good experimental design

Experimental design requires a great deal of planning

Statistics inform us about:

- Mathematical attributes about our data sets
- How data sets relate to each other
- The probability that our claims are correct

However, we need to use human judgement to interpret and understand the results!

Human-generated summary

If we are designing for a widely used application (e.g., Firefox, Brave, or Google Chrome), then perhaps a Type 2 error (less efficiency, but more familiarity) is preferable to a Type 1 error (make users learn a new thing with no performance benefit)

Yet, if we are designing a niche application for expert users, and the tasks involve extremely frequent menu selections (e.g., AutoCAD or OmniGraffle), then perhaps a Type 1 error (more efficiency, at the cost of having to learn a new interface) is preferable to a Type 2 error

EOF

Exam Sample Item

Category “Knowledge”

Which experimental design is more sensitive to order effects?

A Within-subjects

B Randomization

C Between-subjects

D Counter-balancing

Exam Sample Item

Category “Knowledge”

Which experimental design is more sensitive to order effects?

☑ A	Within-subjects
B	Randomization
C	Between-subjects
D	Counter-balancing

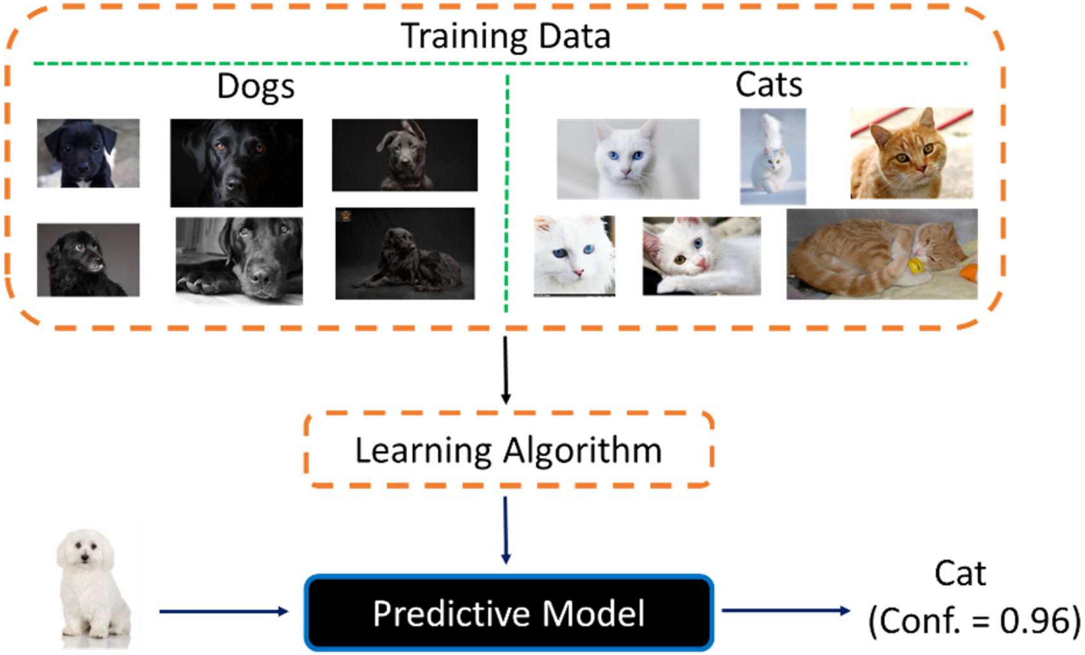
- Randomization and counter-balancing are ways to mitigate the effects of order in task execution

Exam sample item

Category “Insight” Open Question

If the training data in Lakkaraju et al. (2017) are taken as an experimental design,

A) Then what are the two conditions?
B) What is the confound if with a confidence of 96%, the predictive model states that ‘white dogs are cats too?’
C) Although the statement in B is so easy to falsify, then why does not the network do so?



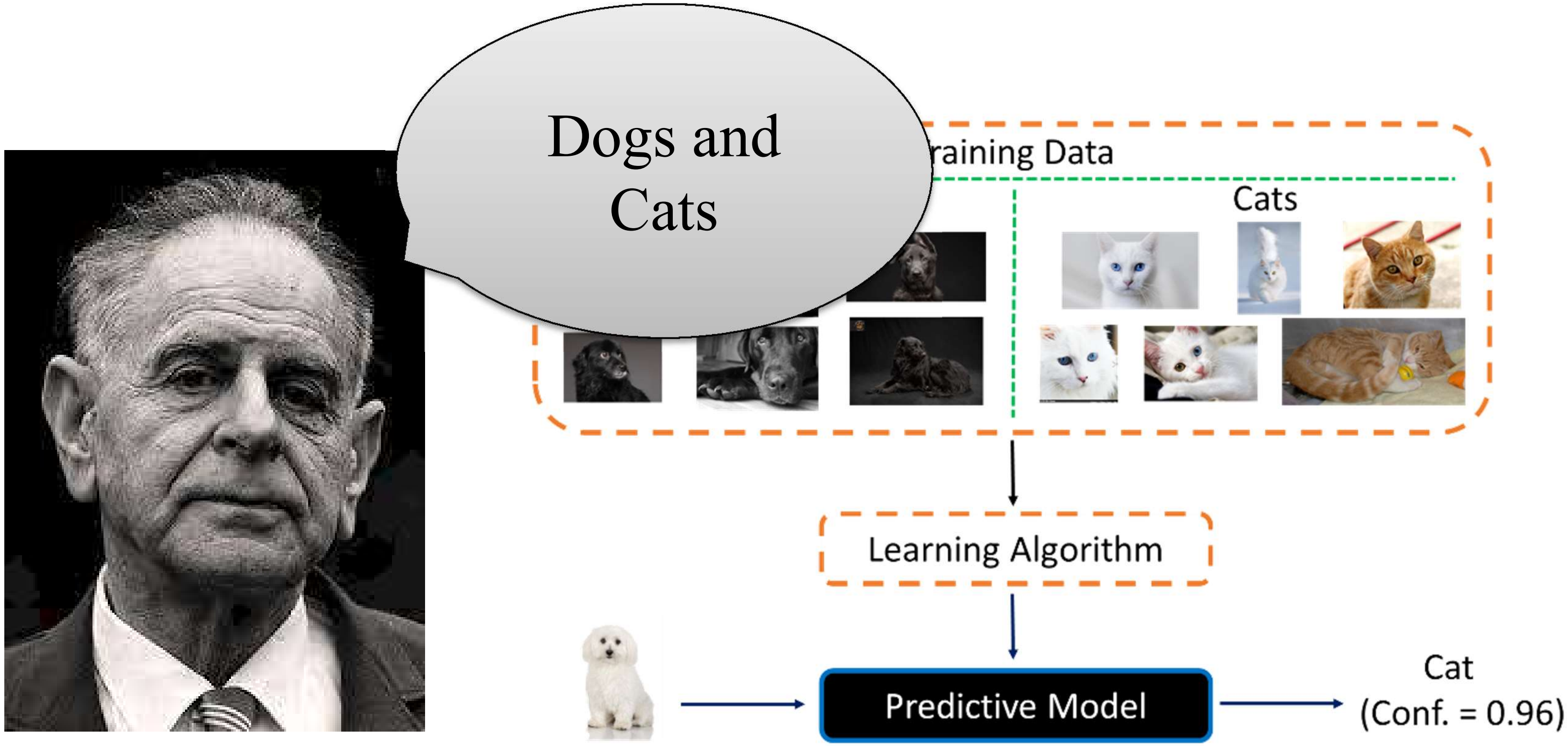
Lakkaraju, H., Kamar, E., Caruana, R., & Horvitz, E. (2017). Identifying unknown unknowns in the open world: Representations and policies for guided exploration. In S. Singh & S. Markovitch (Eds.), *Proceedings of the 31st AAAI Conference on Artificial Intelligence, Feb. 4-7, 2017, SF, CA* (pp. 2124–2132). Palo Alto, CA: AAAI.

Exam sample item

Category “Insight” Open Question

If the training data in Lakkaraju et al. (2017) are taken as an experimental design,

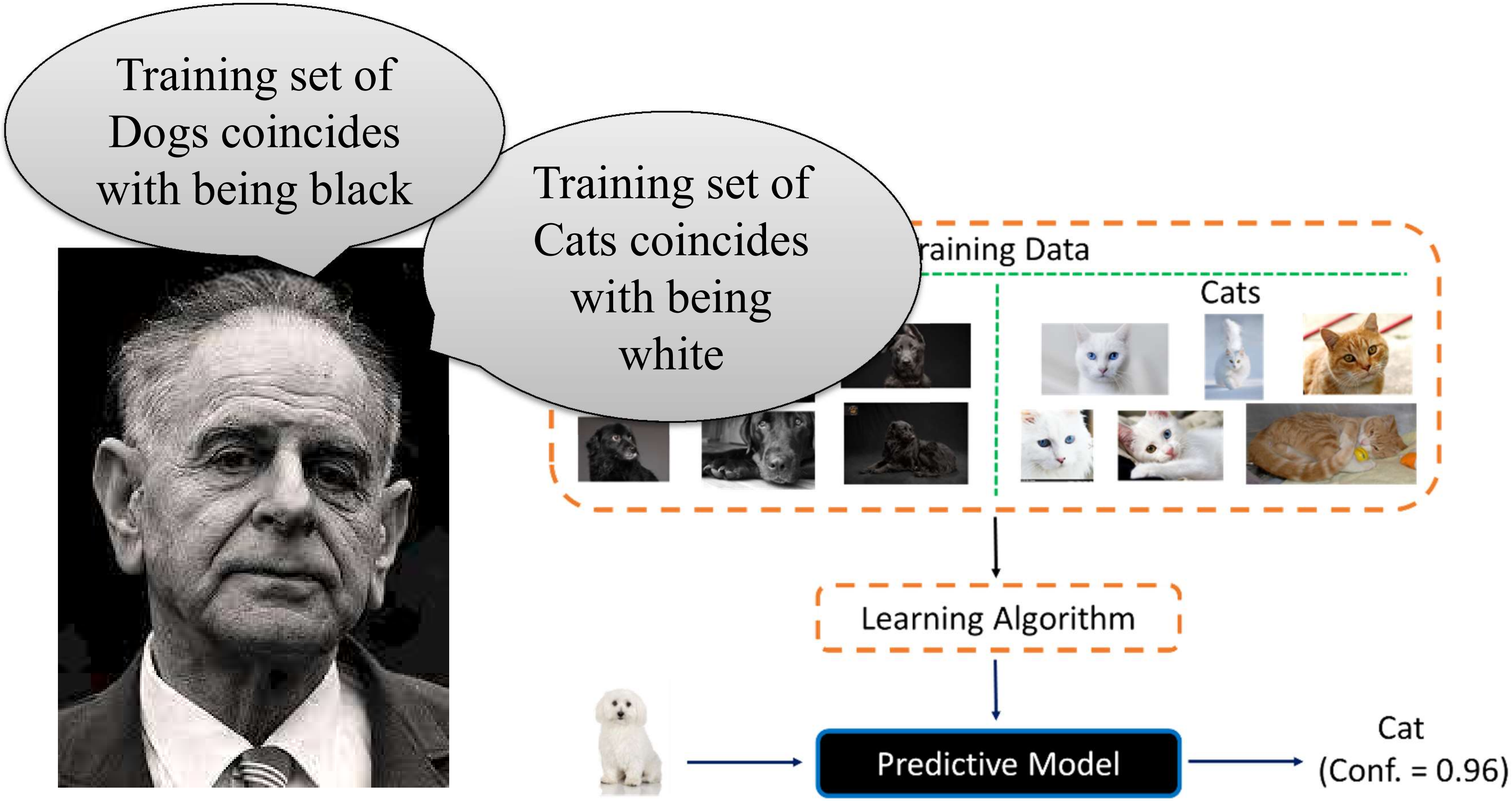
A) Then what are the two conditions?



Exam sample item

Category “Insight” Open Question

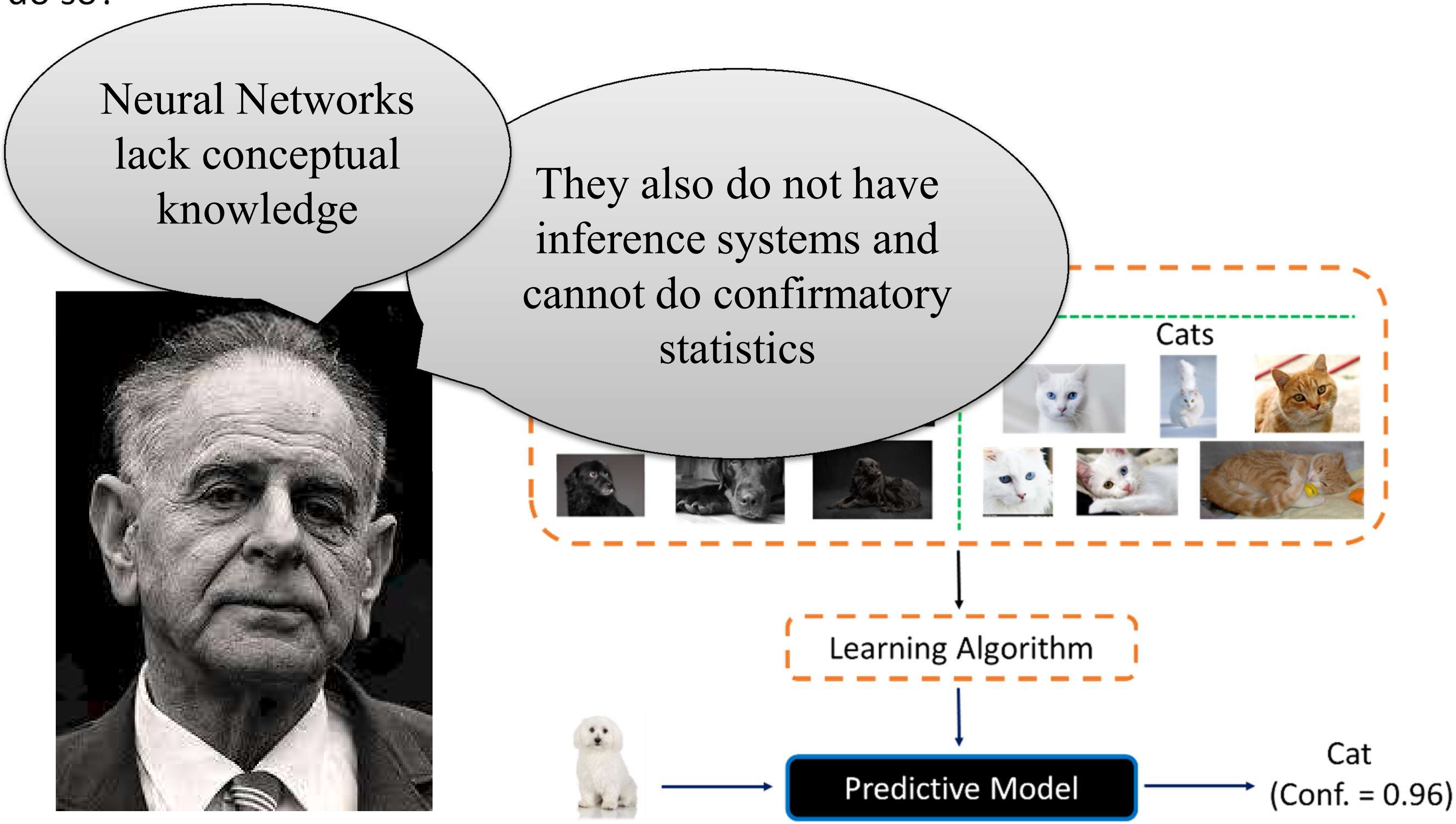
B) What is the confound if with a confidence of 96%, the predictive model states that ‘a white dog is a cat too?’



Exam sample item

Category “Insight” Open Question

C) Although the statement in B is so easy to falsify, then why does not the network do so?



Exam Sample Item

Category “Insight”

If t = 63.657 is minimally required to establish a significant effect, this means that:

A The degrees of freedom must be few

B The rejection area must be small

C α is large

D Both A and B may play a role

Exam Sample Item

Category “Insight”

If t = 63.657 is minimally required to establish a significant effect, this means that:

A The degrees of freedom must be few

B The rejection area must be small

C α is large

D Both A and B may play a role

Table of significance for a two-tailed *t*-test

Rejection area or α set to 1% chance we may be wrong

df	0.2	0.1	0.05	0.02	0.01
1	3.078	6.314	12.706	31.821	$\boxed{63.657}$
2			4.303	6.965	9.925
3			3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.44	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.86	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.25
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.35	1.771	2.16	2.65	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.12	2.583	2.921
17	1.333	1.74	2.11	2.567	2.898
18	1.33	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845

Merely two observations

References

Aron, A., & Aron, E. N. (1999). *Statistics for psychology.* Prentice-Hall, Inc.

McGrath, J. E. (1995). Methodology matters: Doing research in the behavioral and social sciences. In R. M. Baecker, et al. (Eds.), *Readings in Human–Computer Interaction* (pp. 152-169). Morgan Kaufmann.

Student (1908). The probable error of a mean. *Biometrika, 6*(1), 1-25.

NEXT WEEK

Response session: Two days in advance, you write to me the issues that you still have, which I will explain again in class. That means you already studied everything so that your questions are specific and informed

This is the time to ask what is main content and what are asides. What is regarded as really important and what are little sidesteps. Don't miss that opportunity

We will also do a test exam to let you get used to the style of questions